



29th August 2025

Xiang Li: JRF 2025

Dear Committee,

It is my great pleasure to write this recommendation letter for Xiang Li, who is a PhD student in my research group.

Xiang started his PhD in Michaelmas 2022 and his PhD submission deadline is 30th September 2026. Xiang was admitted through a highly competitive process, as illustrated by his China Scholarship Council International Cambridge Scholarship.

Xiang's PhD is in the field of virtual and augmented reality. Specifically, he investigates new interaction techniques that enable people to interact in such environments in an effective, efficient, and safe manner. As part of this work, he has been exploring *swarm manipulation* as a novel and promising interaction technique. Briefly, this draws inspiration from physical swarm particle control, which in virtual environments can be utilized without the usual drawbacks, in particular gravity, inertia, and stabilization and control issues. Xiang's first paper in his PhD, which he proposed and first-authored, *Swarm manipulation in virtual reality*, explores this idea by implementing a virtual swarm particle hand that provides significant benefits in 3D object manipulation tasks. This work was published at the 11th ACM Symposium on Spatial User Interaction (SUI 2023), which is an internationally leading publication venue in this area. This paper was later selected for a special journal issue which allowed Xiang has expanded on this work and was later published in the journal *Computers & Graphics* (2025).

In addition, he has a first-authored paper on on-body menus in virtual reality published in 2024 at the *Proceedings of the ACM on Human-Computer Interaction* (as a special issue of *ACM Conference on Interactive Surfaces and Spaces* conference). This year he has a first-authored paper in 13th ACM Symposium on Spatial User Interaction (SUI 2025), which was selected as the University of Cambridge's representative at Stockholm International Youth Science Seminar 2025, during Nobel Week in Stockholm, Sweden this year. In addition, he has another co-authored paper which will appear in the top journal *IEEE Transactions on Visualization and Computer Graphics*. He also has a first-authored paper under minor revision in *IEEE Transactions on Visualization and Computer Graphics*. The latter paper on micro gestures for text editing in immersive environment won the 2025 UKRI Innovate UK Immersive Tech Award for Technical Innovation this year.

Xiang is a highly creative researcher who is particularly excellent at ideating high-quality research ideas and turning them into reality. On a personal level, Xiang is a very approachable person, and he is friendly, helpful and diligent. In addition, he has significant teaching experience, for example, he regularly supervises in human-computer interaction, interaction design and part II projects for the Department of Computer Science and Technology at the University of Cambridge. I have no doubt he will emerge as a leading researcher in his field.

Based on Xiang's profile and prior research he is excellently suitable for a JRF and I give him my highest recommendation.

Yours sincerely,

Professor Per Ola Kristensson, PhD



Bio:

Per Ola Kristensson is a Professor of Interactive Systems Engineering (Personal Chair) in the Department of Engineering at the University of Cambridge and a Fellow of Trinity College, Cambridge. He is a co-founder and co-director of the Centre for Human-Inspired Artificial Intelligence at the University of Cambridge. He is interested in designing intelligent interactive systems that enable people to be more creative, expressive and satisfied in their daily lives. His PhD thesis was on gesture keyboard technology, which he co-invented together with Dr Shumin Zhai in 2001-2002, and in 2007 he co-founded ShapeWriter, Inc. to commercialise this technology. He was the Director of Engineering of this company until it was acquired by Nuance Communications in 2010. ShapeWriter was selected as the 8th best iPhone application by Time magazine in 2008 and won a Google Android ADC50 developer award in the same year. Today gesture keyboard technology, sometimes called gesture typing, swyping or QuickPath, is a ubiquitous mobile text entry method available in most mobile phones and tablets and is currently making its way into emerging technologies, such as virtual and augmented reality headsets. He did his doctoral work at the Institute of Technology at Linköping University, Sweden and at IBM Almaden Research Center in San Jose, California, USA (Ph.D. Computer Science 2007). In 2008-2011 he was a Junior Research Fellow at the University of Cambridge (Darwin College) and in 2011-2014 he was a Lecturer at the University of St Andrews. He is an Honorary Associate Professor (Docent) in Computer and Systems Science at Stockholm University, Sweden and was an Honorary Reader at the University of St Andrews (2014-2017). In 2013 he was recognised as an Innovator Under 35 (TR35) by MIT Technology Review and appointed a Member of the Royal Society of Edinburgh Young Academy of Scotland. In 2014 he won the ACM User Interface Software and Technology (UIST) Lasting Impact Award and the Royal Society of Edinburgh Early Career Prize in Physical Sciences, the Sir Thomas Makdougall Brisbane Medal. He is an Associate Editor of ACM Transactions on Computer-Human Interaction and ACM Transactions on Interactive Intelligent Systems and served as a Technical Program Co-Chair for ACM CHI 2023, the premier international conference on human-computer interaction.